Design Verification Plan and Report (virtual-test version) — VBF-T1R2-DVPR-01

Case t1_r2; nominated design V4b. All results below are simulation results (PyBaMM SPMe/DFN + lumped thermal + zero-D thermal runaway); physical-test items are honestly marked N/A.

#	Verification item	Condition	Result value	Determination	Source
1	1C discharge capacity	\`run-pyamm --protocol 1C_discharge\`, 298.15 K	5.6997 Ah	PASS (≥ nominal 5.0 Ah)	\`cell/r4_v4b_1c_spme.json:capacity_ah\`
2	Voltage window	parameter set cut-offs	2.5 – 4.2 V	PASS (design target; overcharge limit 4.7 V = upper + 0.5 V protocol)	parameter set \`Lower/Upper voltage cut-off [V]\`
3	4C fast-charge temperature rise	\`run-pyamm --protocol 4C_charge_45C\` equivalent — DFN, lumped thermal, h = 60 W/m²·K, 298.15 K ambient	T_max 326.46 K (53.31 °C)	PASS (≤ 333.15 K; margin 6.69 K)	\`cell/r4_v4b_4c_dfn.json:T_max_K\`
4	4C fast-charge plating	same run; negative electrode potential vs Li/Li	min +0.0215 V → plated = false	PASS (≥ 0 V required)	\`cell/r4_v4b_4c_dfn.json:anode_potential_v\` (min, mechanical)
5	Energy density	\`bda calc-energy\` contract formula (∫V·I dt ÷ Σ layer mass; electrolyte excluded)	583.2 Wh/kg	PASS (≥ 392.61 Wh/kg)	\`cell/r4_v4b_energy.json:energy_density_wh_kg\`
6	Overcharge to 4.7 V → thermal runaway	\`run-pyamm --protocol overcharge\` (0.5C to 4.7 V) + \`bda run-tr\` (mcp 31.72 J/K = mass × 900; hA 0.3186 = 60 × 0.00531 design-consistent)	T_max 299.43 K; triggered = false	PASS (no thermal runaway; note: lumped 0-D proxy caliber — physical validation requires experiment)	\`cell/r4_v4b_overcharge.json:T_max_K\`; \`cell/r4_v4b_tr.json:triggered\`
7	Fast-charge CC acceptance (informational)	4C charge to 4.2 V, CC-only	4.53 Ah (79% of 1C capacity)	INFO (no criterion; documents the voltage-limited stop)	\`inferred\`: (t_end − t_vmin) × 20 A / 3600, from \`cell/r4_v4b_4c_dfn.json\` voltage curve

Not covered (N/A, beyond pure simulation boundary — requires physical experiment):

- Nail penetration — N/A
- Crush — N/A
- Drop / mechanical shock — N/A
- Cycle life (aging) — N/A for this case (durability not in criteria; Chen2020 set is aging-capable, protocol available for follow-up)
- Rate-pulse internal resistance (HPPC) — N/A; DCR 80.4 μΩ available as 1C-point estimate (\`cell/r4_v4b_energy.json:dcr_ohm\`)
- True DFT/MD molecular endorsement — skipped by design (entry-0 meta \`real_compute: false\`; endorse entry in log.jsonl records the skip honestly)

Conclusion

All simulation-verifiable items PASS against the entry-0 criteria (mechanical `bda log-evaluate` verdicts, round 4, in `log.jsonl`). Uncovered items listed above are cited as paper limitations.